

The Duality of Logic: Unifying Discovery and Invention

—A Meta-Theoretical Solution Based on Systemic Ontology

Author: Guorui He (Independent Researcher)

Email: zhicology@qq.com

ORCID: 0009-0006-2947-0032

Abstract

What is logic? This question has haunted philosophy for over two millennia. One camp holds that logic reflects the objective structure of the world—humans merely *discover* it. The other contends that logic is nothing more than a set of arbitrary symbolic conventions—we *invent* it. Both positions have merit, yet the debate remains stalemated because they share an unexamined premise: logic must be *either* discovered *or* invented.

This paper aims to break the stalemate. We introduce the framework of *duality* from systemic ontology as a working hypothesis and conduct an axiomatic deduction by substituting "the logical system" as the object of analysis. The result is a unified account: logic simultaneously possesses a *facet of discovery* and a *facet of invention*. These are not mutually exclusive alternatives but two inseparable facets of a single system.

The facet of discovery is anchored in the meta-rules of the universe—the laws of non-contradiction, identity, and causality are constitutive conditions for any stably assertable system. They are independent of human will and endow logic with its cross-cultural normative force. Humans can only *discover* them.

The facet of invention manifests as symbolic systems—propositional forms, inference rules, truth tables, and formal systems are cognitive interfaces and *noetic genes* created by humans to represent these meta-rules. Shaped by the path-dependence of cognitive frameworks, they evolve with civilization. Humans *invent* and continually revise them.

Without the facet of discovery, logic would degenerate into arbitrary symbolic play; without the facet of invention, logic could not enter public cognition or evolve. Only the unity of both facets constitutes a complete logical system.

This framework not only dissolves the "discovery *versus* invention" dichotomy but also yields two testable corollaries: (1) an ecological dynamics of logical evolution—the transformation of logical symbol systems is driven by selection pressures toward more efficient representation of meta-rules; and (2) a duality principle for AI logic design—the logical modules of artificial intelligence should possess both an immutable core anchored in meta-rules (facet of discovery) and a learnable, extensible symbolic interface (facet of invention).

The argument of this paper itself constitutes a meta-theoretical self-reference loop: we use logical tools to demonstrate the nature of logic and entrust the power of adjudication to the reader as "external adjudicator." Thus, this paper not only advances a new theory of logic but also demonstrates how meta-theoretical discourse can achieve self-legitimation without falling into paradox—a contribution that may prove to be its most enduring.

Keywords: philosophy of logic; duality; discovery and invention; meta-theory; self-reference; AI logic design

Glossary of Core Terms

To facilitate understanding of the core argument, the following table provides advance clarification of several specialized terms that appear frequently throughout this paper, along with their approximate correspondence to existing academic concepts.

Term	Concise Definition	Correspondence to Existing Concepts
Duality	The property of a system simultaneously manifesting as two inseparable facets of existence. In this paper, it refers specifically to the dual nature of logical systems as possessing both a "facet of discovery" (the mirroring of meta-rules) and a "facet of invention" (noetic genes).	Borrows metaphorically from "wave-particle duality" in physics, though its connotations are more properly ontological. It bears affinity to suggestions of logical dualism in Quine's thesis of the revisability of logic, but provides an explicit deductive framework.
Facet of Discovery	The facet of logical duality anchored in objective meta-rules. The fundamental laws of logic (non-contradiction, identity, causality) are constitutive conditions that any stably assertable system must satisfy, independent of human will.	Corresponds to the realist position in philosophy of logic, though this paper treats it not as an opposing alternative but as one pole of a unified whole.
Facet of Invention	The facet of logical duality shaped by human cognitive frameworks and historical path-dependence. Symbolic systems (propositional forms, inference rules, formal systems) are cognitive interfaces created by humans that evolve with civilization.	Corresponds to the conventionalist position in philosophy of logic, likewise treated as one pole of a unified whole rather than an independent thesis.
Meta-Rule	A constitutive condition more fundamental than specific physical laws or social norms, which any stably existing system must satisfy. This paper argues that non-contradiction, identity, and causality are meta-rules.	Functionally similar to Kantian "a priori categories," though without presupposing a transcendental subject; closer to "systemic conditions of persistence" in systems theory.
Noetic Gene	A unit of information capable of replication, variation, and selection within cognitive systems. Logical	Corresponds closely to Dawkins' concept of the meme, but situated within the framework of systemic

Term	Concise Definition	Correspondence to Existing Concepts
	symbol systems, conceptual frameworks, and narrative structures are all noetic genes.	ontology, emphasizing its dual unity with a "material carrier."
Self-Reference	The capacity or phenomenon of a system taking itself or its own processes as an object of cognition or operation. In this paper, it refers both to the ability of logical systems to describe themselves (e.g., Gödel numbering) and to this paper's own reflection on the logical tools it employs.	Corresponds to self-reference in logic, reflexivity in philosophy, and "observing oneself" in second-order cybernetics.
External Adjudication	The mechanism of meta-theoretical legitimacy proposed in this paper: the ultimate legitimacy of a meta-theoretical argument derives not from circular self-justification within the system, but from adjudication by the reader as an external observer, based on the argument's logical coherence and explanatory power.	Structurally isomorphic to the insight in Gödel's incompleteness theorems that "truth is seen from outside the system"; methodologically akin to the independent testing of theories by evidence in philosophy of science.
Meta-Theory	A theory that takes theory itself as its object of study. This paper is both a theory <i>of</i> logic and a theory <i>about</i> how to legitimately speak of logic.	Standard academic usage.

Note: The "Correspondence to Existing Concepts" column is intended to provide cognitive anchors for readers familiar with relevant fields; it does not claim that the terms in this paper are strictly equivalent to the concepts listed. The precise meanings of terms in this paper are determined by their definitions in the main text.

1. Introduction: The Millennial Debate on the Nature of Logic

1.1 Statement of the Problem

What is logic? This seemingly simple question has troubled human thought for over two millennia. It concerns our understanding of the nature of reason, the foundations of truth, and the very possibility of knowledge.

In the history of philosophy, two fundamentally opposed positions have emerged in response to this question.

One side maintains that logic is **discovered**. Plato held that the truths of logic and mathematics reside in an eternal realm of Forms, and that human reason merely recollects and uncovers them. The early Russell and Gödel held similar views: logical propositions describe the most general structure of existence, independent of human cognitive activity. In this sense, logic is like the American continent—it was always there, awaiting its Columbus to discover it.

The other side maintains that logic is **invented**. The later Wittgenstein regarded logic as the rules of a "language-game"—a set of symbolic conventions we establish by agreement. Carnap stated even more explicitly that logical rules are chosen by humans, and the criterion for choice is not "truth" but "convenience." Quine, with his thesis of the "revisability of logic," pushed this position to its extreme: if experience conflicts with theory, we may even consider revising logic itself. In this sense, logic is like the rules of chess—invented by humans, agreed upon by humans, and revisable by humans.

Both positions possess undeniable rationales. Discovery-ism can explain the *compulsory force* of logic: Why is the law of non-contradiction valid across all cultures? Because it is a structure of objective reality, not subject to our will. Yet it struggles to explain the *diversity* of logic: If logic is discovered, why do there exist multiple different logical systems—classical logic, intuitionistic logic, fuzzy logic, and so on? Why have logical symbols and inference rules undergone such conspicuous historical transformations?

Invention-ism fares conversely. It can explain the diversity and historical evolution of logic: different logical systems are like different languages or game rules, each suited to its own context. Yet it struggles to explain the compulsory force of logic: If logic is merely our convention, why can we not convene a logic that permits "A and not-A simultaneously" and use it to think effectively about the world? Why do certain logical laws exhibit cross-cultural and transhistorical universality?

1.2 The Nature of the Impasse

The diagnosis offered in this paper is that this debate has remained unresolved for millennia not because of insufficient evidence or imprecise argumentation, but because both sides share an unexamined premise—that logic must be *either* discovered *or* invented.

This "either-or" framework is itself the root of the problem. It forces us to select one facet of an incomplete picture and then deny the existence of the other. Discovery-ists see the normative foundation of logic but cannot account for the diversity of symbolic systems; invention-ists see the artificiality of symbolic systems but must evade the question of where normativity comes from.

If we can step outside this either-or framework, might we obtain a new perspective—one that can accommodate both insights simultaneously?

1.3 The Proposed Solution

The answer offered in this paper is affirmative. We introduce a theoretical framework derived from systemic ontology—**duality**.

The core claim of duality is that any system attaining a certain degree of complexity simultaneously manifests as two inseparable modes of existence—**the material mode** (carrier and energetic foundation) and **the noetic mode** (patterns and rules of informational organization). The hardware and software of a computer, the DNA molecule of life and the genetic information it encodes, the material infrastructure of a society and its institutional culture—these are all instances of duality. The two modes are not mutually exclusive alternatives but two facets of a single system: there is no rule without a carrier, and no carrier without a rule.

This paper will argue that the logical system likewise exhibits duality.

Specifically, logic simultaneously manifests a dual existence as a **facet of discovery** and a **facet of invention**:

- **Facet of Discovery** (corresponding to the material mode): The fundamental laws of logic—the laws of non-contradiction, identity, and causality—are stable structural mirrorings of cosmic meta-rules within human cognitive frameworks. They are constitutive conditions that any stably assertable system must satisfy, independent of human will. Humans can only **discover** them and cannot alter them at will.
- **Facet of Invention** (corresponding to the noetic mode): The symbolic systems of logic—propositional forms, inference rules, truth tables, and formal systems—are cognitive interfaces and **noetic genes** created by humans to describe the aforementioned meta-rules. Their forms are shaped by the path-dependence of cognitive frameworks and evolve with civilization. Humans **invent** and continuously modify these symbolic shells.

The two facets are inseparable: without the facet of discovery, logic would degenerate into arbitrary symbolic play, losing its normativity and predictive power; without the facet of invention, logic could not enter public human cognition, nor be transmitted, taught, or revised.

The distinctive contribution of this framework is this: it does not seek a "middle ground" between discovery-ism and invention-ism, but directly dissolves the "either-or" mode of questioning. Discovery-ists are right—logic does possess a foundation independent of human will; invention-ists are also right—logical symbolic systems are indeed human creations that evolve over history. Each has glimpsed one facet of the duality of logic, but mistaken that facet for the whole.

1.4 Roadmap of the Argument

The argument of this paper proceeds in six steps, forming a complete structure from premise, through deduction and elaboration, to a reflexive closure.

Step One (Section 2) introduces duality as a working hypothesis. This paper does not require the reader to accept the entire systemic ontology in advance. Instead, duality is introduced as a testable analytical framework: if the logical system exhibits duality, what phenomena should we be able to observe? The explanatory power of this hypothesis will be underwritten by the subsequent deduction and corollaries.

Step Two (Section 3) substitutes "the logical system" into the duality framework and conducts an axiomatic deduction. Proceeding from the working hypothesis of duality, we derive that logic necessarily possesses a facet of discovery and a facet of invention, and demonstrate that the two facets are inseparable. This is the logical axis of the entire paper.

Step Three (Sections 4–5) elaborates the concrete content of the facet of discovery and the facet of

invention respectively. Section 4 argues why the laws of non-contradiction, identity, and causality constitute the mirroring of meta-rules, providing an ontological foundation for the normativity of logic. Section 5, drawing on cases from the history of logic, shows how logical symbol systems, as noetic genes, undergo variation and selection.

Step Four (Section 6) derives two corollaries from the theorem—the ecological dynamics of logical evolution and the duality principle for AI logic design—to demonstrate that this theorem is not empty speculation but can generate testable empirical predictions and actionable engineering principles.

Step Five (Section 7) undertakes a meta-reflection to complete the paper's "self-reference loop." This paper takes its own argument as an object, examining whether it too obeys duality, and by introducing the mechanism of "external adjudication," answers a question that no meta-theory can evade: How can a theory about logic legitimately use logic to argue for itself?

Step Six (Section 8) concludes the paper and indicates directions for future research.

2. Theoretical Foundations: Duality as a Working Hypothesis

Before unfolding the core argument, it is necessary to clarify the content and provenance of the foundational framework employed in this paper—"duality"—and to specify its methodological status within the argument.

2.1 The Conceptual Genealogy of Duality

The term "duality" borrows metaphorically from the "wave-particle duality" of physics, but its connotations in this theoretical framework are more fundamental. Wave-particle duality describes two complementary manifestations of a single physical entity under measurement; the duality discussed in this paper refers to the dual constitution of any complex system at the ontological level. It is not a matter of two "appearances" of a single entity under observation, but of two inseparable constitutive facets through which the system exists.

This concept originates in the author's theoretical system, *Systemic Ontology*. Within that system, duality is formulated as a universal systemic axiom:

Axiom 2 (Material/Noetic Duality): Any system that attains a certain degree of complexity simultaneously manifests as a dual existence of material entity and noetic pattern. The material mode is the carrier and energetic foundation of the system; the noetic mode is the pattern, rules, and representation of informational organization within and between systems. The two are mutually entailing and inseparable.

Three cross-domain examples may serve to clarify its content:

Example 1: The Computer System. A computer consists simultaneously of hardware (material mode) and software (noetic mode). The hardware comprises silicon chips, circuits, and memory—the physical carrier; the software comprises the operating system, applications, and algorithms—the informational pattern. Hardware without software is merely inert silicon; software without hardware cannot execute. The two do not arise sequentially but jointly constitute the system "computer." The laws governing software (e.g., algorithmic complexity, user experience) cannot be reduced to the switching laws of transistors, yet they must take hardware as their substrate.

Example 2: The Living System. An organism consists simultaneously of a material carrier and genetic information. The DNA molecule as a material entity (material mode) has a double-helix structure that

encodes the genetic information (noetic mode) governing biological development and metabolism. The chemical decomposition of DNA and informational mutation obey laws at different levels: the former follows chemical laws, the latter evolutionary dynamics. Without the DNA molecule, genetic information cannot be instantiated in the physical world; without genetic information, DNA is merely a collection of carbon-based macromolecules.

Example 3: The Social System. A society consists simultaneously of a material base and an institutional culture. Population, land, infrastructure, and technological equipment constitute the material mode of society; laws, institutions, customs, language, and shared narratives constitute the noetic mode. Institutional change (e.g., from monarchy to democracy) cannot be reduced to fluctuations in population or the replacement of buildings, yet institutions must take material carriers as their substrate in order to operate. An institution without a material base is mere empty text; a population without institutional organization is merely a disorganized crowd.

These three examples reveal several core features of duality, which we shall now systematically articulate.

2.2 Core Features of Duality

From the above examples, three core features of duality can be extracted. They are crucial for the subsequent application of the duality framework to the logical system.

Feature One: Inseparability. The material mode and the noetic mode are two facets of a single system. There is no rule without a carrier, and no carrier without a rule. This is not a matter of functional "coordination" but of ontological "co-constitution"—the system is a system precisely because these two facets are simultaneously present. If one facet is absent, the system disintegrates.

Feature Two: Non-Reducibility. The laws of the noetic mode cannot be fully reduced to the laws of motion of the material mode. The operational logic of software cannot be reduced to the equations of electron motion; the patterns of cultural change cannot be reduced to demographic statistical fluctuations. This is not mysticism but a consequence of "emergence": when material components are organized according to a specific noetic pattern, a new level of causal efficacy emerges, and the laws of this level require conceptual tools appropriate to their own description.

Feature Three: Synchronous Emergence. A system and its internal rules emerge synchronously; there is no logical priority between them. The hardware and software of a computer are jointly specified in the same process of system design; life and the genetic code are jointly forged in the same process of biogenesis. There is no "first there is the material carrier, and then the rules 'take effect,'" nor "first there are abstract rules, and then matter 'instantiates' them." System and rule are two facets of a single emergent event.

These three features jointly entail a methodological consequence: **to understand any complex system, one must examine both its material mode and its noetic mode, and one cannot reduce one to the other.** This is the methodological principle that will guide the application of the duality framework to the logical system in this paper.

2.3 Methodology of This Paper: Axiomatic Deduction

Before entering the core argument, the methodological status of the duality framework within this paper must be clarified. This is crucial, for it directly concerns the boundaries of the argument's legitimacy.

This paper does not require the reader to accept the entire axiomatic system of *Systemic*

Ontology in advance. Rather, it adopts a more modest and dialogical strategy: duality is introduced as a **working hypothesis**.

By "working hypothesis," we mean the following:

- **Independence of the hypothesis:** Duality is provisionally posited as an analytical framework. Its validity is underwritten not by the overall authority of its source system, but by its deductive fruits within this paper—namely, its explanatory power with respect to the duality of logic.
- **Operability of the hypothesis:** Duality is not a vague philosophical metaphor but an analytical tool with clearly specified features (inseparability, non-reducibility, synchronous emergence). It can guide us in structurally decomposing the object "the logical system."
- **Testability of the hypothesis:** From the duality hypothesis, this paper derives two concrete corollaries (the ecological dynamics of logical evolution, the duality principle for AI logic design). Whether these corollaries hold constitutes an indirect test of the original hypothesis. If a corollary is falsified, the hypothesis must be revised or abandoned.

The substance of this methodological strategy is this: **to transform an internal deduction of an axiomatic system into a hypothesis-testing structure capable of external academic dialogue**. The reader need not affirm the ontological status of duality; they need only acknowledge that it is an analytical framework worth taking seriously, and judge its efficacy based on the deductions and evidence presented in this paper.

2.4 From Duality to the Logical System: The Starting Point of the Argument

We now possess all the tools necessary to enter the core argument.

The object of analysis in this paper is the **logical system**—a complex system constituted by fundamental laws (non-contradiction, identity, causality, etc.), symbolic systems (propositional forms, inference rules, truth tables, etc.), semantic interpretations, and practical applications.

According to the working hypothesis of duality, any system attaining a certain degree of complexity simultaneously manifests the dual existence of a material mode and a noetic mode. The logical system unquestionably meets the threshold condition of complexity: it contains multi-level structure, possesses the capacity for self-reference (e.g., Gödel numbering), and performs irreplaceable systemic functions in human cognition and civilizational evolution.

Therefore, we have ample reason to substitute the logical system into the duality framework and ask:

- **Question One:** What corresponds to the "material mode" of the logical system? It must be the normative skeleton that enables the system to exist—namely, the rules themselves, independent of any particular symbolic representation.
- **Question Two:** What corresponds to the "noetic mode" of the logical system? It must be the representational forms that enable the rules to be processed, transmitted, and evolved by human cognitive systems—namely, symbolic systems with symbols as their stable carriers.
- **Question Three:** Do these two dimensions satisfy the core features of duality—inseparability, non-reducibility, and synchronous emergence?

The following sections will answer the first two questions in turn; the third will be addressed in Section 3.5.

This line of inquiry will lead directly to the core theorem of this paper—the Duality Theorem of Logic.

3. Core Argument: The Duality Theorem of Logic

Before commencing the deduction, it is necessary to provide a clear demarcation of the object of analysis—"the logical system."

As understood in this paper, "the logical system" designates a complex system constituted by the following four levels:

Level One: Fundamental Laws. This is the most stable and universal kernel of the logical system. It includes the law of non-contradiction ($\neg(A \wedge \neg A)$), the law of identity ($A = A$), the law of excluded middle ($A \vee \neg A$), and the fundamental framework of causal reasoning (if A is a sufficient condition for B, then B can be inferred from A). These laws constitute the normative baseline for any valid reasoning.

Level Two: Symbolic Systems. These are the formal tools employed within the logical system to express and manipulate the above laws. They include propositional variables, logical connectives ($\neg$, $\wedge$, $\vee$, $\rightarrow$, $\leftrightarrow$), quantifiers ($\forall$, $\exists$), inference rules (such as Modus Ponens), truth tables, proof systems, and so forth. Different logical traditions possess different symbolic systems.

Level Three: Semantic Interpretation. These are the mapping rules that connect the symbolic system to the external world or to possible worlds. They include truth conditions, model-theoretic semantics, proof-theoretic semantics, and the like.

Level Four: Practical Application. This is the actual deployment of the logical system in human cognitive activities. It includes everyday reasoning, scientific argumentation, mathematical proof, legal debate, program design, and so on.

When we ask "Is logic discovered or invented?", we are in effect asking: among these four levels, which are "discovered" and which are "invented"? This paper will argue that Level One belongs to the facet of discovery, and Level Two belongs to the facet of invention. Levels Three and Four are the coupled products of the two facets in specific contexts.

3.2 Substituting into the Duality Framework

According to the working hypothesis of duality established in Section 2:

Any system that attains a certain degree of complexity simultaneously manifests the dual existence of a material mode and a noetic mode. The material mode is the carrier and energetic foundation of the system; the noetic mode is the pattern, rules, and representation of informational organization within and between systems.

The logical system unquestionably meets the threshold condition of "a certain degree of complexity": it contains multi-level structure, possesses the capacity for self-reference (e.g., Gödel numbering), and performs irreplaceable systemic functions in human cognitive ecology.

Therefore, we have ample reason to substitute the logical system into the duality framework. The substitution yields the following questions:

Question One: What corresponds to the "material mode" of the logical system? It must be the normative skeleton that enables the system to exist—namely, the **rules as such**, independent of any particular symbolic representation.

Question Two: What corresponds to the "noetic mode" of the logical system? It must be the representational forms that enable the rules to be processed, transmitted, and evolved by human cognitive systems—namely, the **symbolic systems** that take symbols as their stable carriers.

Question Three: Do these two dimensions satisfy the core features of duality—inseparability, non-reducibility, and synchronous emergence?

The following subsections (3.3 and 3.4) will answer the first two questions in turn; the third will be addressed in Subsection 3.5.

3.3 The Facet of Discovery: Rules as Such

We designate the "material mode" of the logical system as the **facet of discovery**. The rationale for this designation is as follows: the fundamental laws of logic—the laws of non-contradiction, identity, and causality—are not the products of human convention, but are rather **rules as such** that any stably assertable system must satisfy. They constitute the normative skeleton of the logical system, independent of any particular symbolic representational form.

The argument proceeds in three steps.

Step One: The validity of the rules is not conferred by symbolic convention.

Consider the law of non-contradiction as an example. We can express it in multiple symbolic forms:

- Aristotelian formulation: "The same attribute cannot at the same time belong and not belong to the same subject in the same respect."
- Propositional logic formulation: $\neg(P \wedge \neg P)$
- Predicate logic formulation: $\forall x \neg(F(x) \wedge \neg F(x))$
- Natural language formulation: "One cannot both be and not be."

These symbolic expressions differ, yet they point to one and the same rule. The validity of this rule does not depend on which symbols we choose to write it down. If the validity of non-contradiction were conferred by symbolic convention, then altering the symbolic convention should be able to alter or abolish it—but this is manifestly not the case. No matter what symbols we use to write it, the normative constraint delineated by non-contradiction remains invariant.

This indicates that **rules as such** and **symbolic representations of rules** belong to two distinct levels. Rules as such belong to the facet of discovery; symbolic representations belong to the facet of invention.

Step Two: The rules are constitutive conditions for the stable existence of a system.

Why does the law of non-contradiction possess this validity independent of symbolic convention? The answer lies in this: non-contradiction is a constitutive condition that any stably assertable system must satisfy.

A system that permits "A and not-A simultaneously true" cannot internally sustain any stable distinction. What is affirmed today can be denied tomorrow; what is deduced can simultaneously be retracted. Such a system logically annihilates itself—it cannot maintain any determinate assertable content, and thus cannot exist as an "assertable system" at all.

This is not the result of human "stipulation," but a constitutive condition of systemic existence itself. Just as a physical system, in order to exist stably, must satisfy constraints such as energy conservation, so too a logical system, in order to exist stably, must satisfy constraints such as non-contradiction. These constraints are not externally imposed norms, but internal determinations of the system's being what it is.

Step Three: Humans discover the rules; they do not invent them.

If non-contradiction, identity, and causality are constitutive conditions for any stably assertable system, then their existence is independent of whether humans recognize them—just as universal gravitation was already operative before Newton.

In the long course of cognitive practice and reflection, humans have progressively made these rules, which are immanent in assertional practice, explicit—moving from "tacit compliance" to "explicit

formulation." This process is one of **discovery**, not invention. Aristotle did not invent the law of non-contradiction; he discovered it—just as Columbus did not invent the American continent, but discovered it.

The fact that different civilizations, in mutual isolation, independently formulated the same fundamental logical laws further supports the thesis of discovery: if the rules were invented, different civilizations should have invented *different* rules; yet they independently *discovered* the *same* rules, because those rules were already immanent in any stable assertional practice.

Conclusion: The facet of discovery of logic is precisely **rules as such**—the fundamental laws of non-contradiction, identity, and causality. They are the "material mode" of the logical system, that is, the normative skeleton that enables the system to exist. Humans can only **discover** these rules; they cannot invent or abolish them. This explains the cross-cultural compulsory force and universal normativity of logic.

3.4 The Facet of Invention: Representational Forms with Symbols as Their Stable Carriers

We designate the "noetic mode" of the logical system as the **facet of invention**. The rationale for this designation is as follows: the symbolic systems of logic—propositional variables, connectives, quantifiers, the formatting of inference rules, truth tables, and so on—are symbolic carriers created by humans in order to represent and manipulate the rules. They are the noetic forms through which the rules are written down, transmitted, taught, and evolved.

The argument proceeds in three steps.

Step One: A single rule admits multiple symbolic representations.

As noted above, the law of non-contradiction can be expressed in Aristotelian subject-predicate form, in propositional truth-functional form, in quantificational predicate-logic form, and in everyday natural language. These symbolic forms differ in expressiveness, conciseness, and operability, yet they represent one and the same rule.

This fact alone demonstrates the independent variability of symbolic systems: the rule remains invariant while the symbols vary. Symbolic systems are not the rules themselves, but **representational forms** of the rules.

Step Two: The artificiality and path-dependence of symbolic systems.

The concrete morphology of logical symbolic systems is shaped by multiple human factors and historical contingencies. This is not a "defect" but a natural feature of the evolution of noetic genes.

- **Linguistic background:** The subject-predicate structure of Aristotelian syllogistic has a deep affinity with the grammatical structure of ancient Greek. In Greek, a typical declarative sentence consists of a subject (*ὑποκείμενον*, "that which lies beneath") and a predicate (*κατηγορούμενον*, "that which is said of the subject"). Aristotle elevated this grammatical structure into a logical category: all propositions can be reduced to the form "S is P," and all reasoning can be reduced to the syllogistic form. This framework was so successful that for two millennia, logic was nearly synonymous with syllogistic. Yet it also imposed profound limitations: relational propositions (e.g., "A is greater than B") are difficult to handle within a subject-predicate structure; the core tools of modern logic—multiple predication, nested quantifiers—cannot be naturally expressed within the syllogistic framework. It was not until Frege, in his 1879 *Begriffsschrift*, replaced the subject-predicate structure with a function-argument structure ($*f^{*}(*x^{*})$) that logic was liberated from the grammatical shackles of natural language. Frege's innovation was not a revision of "rules as

such"—non-contradiction remained just as valid after him as before—but a fundamental reconstruction of the symbolic representational system.

- **Technological medium:** The evolution of logical symbolism has also been profoundly influenced by the media of writing and transmission. Frege's *Begriffsschrift* employed a two-dimensional diagrammatic structure, using horizontal and vertical lines to represent logical relations. This design was conceptually elegant but technologically unsuccessful—the printing technology of the time could not economically reproduce such complex two-dimensional diagrams. In contrast, the linear symbol system adopted by Peano and Russell (e.g., $\supset$ for implication, $\sim$ for negation), though less intuitively perspicuous than Frege's diagrams, was easy to typeset, print, and disseminate. The selective pressure of the technological medium ultimately made the linear symbol system the standard format of twentieth-century logic. This case reveals an important principle: the "victory" of a symbolic system is not purely a matter of logical superiority, but the combined result of logical efficacy, technological feasibility, and ease of dissemination. This is a characteristic feature of noetic gene selection—selective pressures come from multiple dimensions, not a single criterion of truth.
- **Cognitive demands:** The evolution of logical symbolism is also driven by the cognitive demands of different eras. Aristotelian logic served the practices of debate and classification in ancient Greece—syllogistic provided a tool for testing the validity of arguments. Medieval scholastic logic served theological argumentation—the subtle distinctions of term logic (e.g., the distinction between "meaning" and "reference") arose from the need for rigorous analysis of theological propositions. The birth of modern predicate logic was directly driven by the pursuit of rigor in the foundations of mathematics. Frege's goal was to provide a purely logical foundation for arithmetic, which required a symbolic system far more powerful than syllogistic—capable of expressing multiple relations, nested quantifiers, and higher-order predicates. His *Begriffsschrift* was designed precisely for this purpose. The non-classical logics that emerged in the latter half of the twentieth century are even more striking examples of cognitive demands driving symbolic evolution: **fuzzy logic** (invented to handle vague concepts like "tall" or "middle-aged," replacing the two-valued $\{0,1\}$ with the continuous interval $[0,1]$); **intuitionistic logic** (arising from Brouwer's philosophical reflections on mathematical constructivity, rejecting the unconditional applicability of excluded middle over infinite domains); **quantum logic** (proposed to handle non-classical phenomena in quantum mechanics, such as superposition and measurement, modifying the distributive law of classical logic); **non-monotonic logic** (invented to handle commonsense reasoning in artificial intelligence, allowing conclusions to be retracted as new information is added). These "logical variations" are not denials of rules as such—non-contradiction remains in force in some form within fuzzy logic and intuitionistic logic. They are adaptive adjustments at the level of symbolic representation: designing reasoning tools better suited for specific cognitive domains.
- **Historical contingency:** The establishment of certain symbolic conventions is due not to absolute "superiority" but to the choices of key figures, the authority of particular texts, or contingencies of transmission pathways. The now-universal symbols for logical connectives— $\neg$ (negation), $\wedge$ (conjunction), $\vee$ (disjunction), $\rightarrow$ (implication)—were progressively established in the early twentieth century by Peano, Russell, Hilbert, and others.

Before this, Frege had used a completely different set of symbols (e.g., $\vdash$ for assertion, $\supset$ for conditional). The victory of the Peano-Russell system was due partly to its intrinsic merits (conciseness, ease of typesetting), partly to the enormous influence of Russell and Whitehead's *Principia Mathematica*, and partly to the promotion efforts of the Hilbert school. Had history been slightly different—had Frege's *Begriffsschrift* received more attention during his lifetime, or had Peano's system not been taken up by Russell—the symbols in today's logic textbooks might look quite different. This is not to say that the evolution of symbolic systems is entirely arbitrary, but rather that, among multiple functionally equivalent symbolic schemes, the one that ultimately prevails often bears the imprint of historical contingency.

These factors collectively demonstrate that **logical symbolic systems are invented by humans, and their evolution follows the laws of path-dependence.**

Step Three: Symbolic systems as noetic genes.

From an evolutionary perspective, logical symbolic systems are quintessential "**noetic genes**"—units of symbolic information capable of replication, variation, and selection within cognitive systems.

- **Replication:** Logical symbols are transmitted across generations and across cultures through education, texts, and technological media. Every student who learns logic replicates these symbolic patterns and their associated inference rules within their own cognitive system. This replication process is not a simple "copying." In learning $\neg(P \wedge \neg P)$, the student not only memorizes the written form of the symbols but also internalizes the operational rules associated with them (how to read a truth table, how to perform negation and conjunction). This "operational internalization" is a key difference between the replication of noetic genes and biological genes: biological genes transmit only information; noetic genes transmit, along with information, the capacity for information processing.
- **Variation:** Logical symbols undergo variation in transmission. Sources of variation include: **creative innovation** (Frege's invention of quantifier symbols was a fundamental symbolic variation, replacing the cumbersome natural-language "for all x " and "there exists an x " with $\forall x$ and $\exists x$, thereby enabling nested quantifiers and multiple predication); **cross-domain transfer** (Boole transferred algebraic symbolism into the domain of logic, creating Boolean algebra, using $\times$ for conjunction and $+$ for disjunction—a symbolic choice rooted in his analogy with algebra rather than any intrinsic requirement of logic); **adaptive adjustment** (Zadeh, in order to handle vague concepts, extended the truth-value set of classical logic from $\{0,1\}$ to the interval $[0,1]$ and redefined the operational rules of the connectives, e.g., using $\min$ for conjunction—a key variation enabling the symbolic system to adapt to a new cognitive domain); **misreading and reinterpretation** (in the history of logic, "creative misreadings" of predecessors by successors are often a source of variation; Brouwer's rejection of excluded middle can be seen as a "misreading" of the rules of classical logic—proceeding from his philosophical stance of mathematical constructivity, he discerned the limitations of classical logic's applicability over infinite domains).
- **Selection:** Not all symbolic variations survive. Selective pressures come from multiple dimensions: **cognitive efficiency** (can the symbolic system express complex reasoning concisely? A mathematical proposition that modern predicate logic can express in a few short lines might require pages of cumbersome derivation within the syllogistic framework—this efficiency advantage was a key factor in predicate logic's selective advantage); **systemic compatibility** (can the new symbolic system integrate with existing bodies of knowledge?

Fuzzy logic has been widely accepted partly because it is a "conservative extension" of classical logic—when membership degrees are restricted to $\{0,1\}$, fuzzy logic collapses back into classical logic, and this compatibility reduces the acceptance cost for the academic community); **problem-solving power** (can the new symbolic system solve problems that the old system could not effectively handle? Non-monotonic logic has gained traction in AI precisely because classical logic cannot handle "defeasible reasoning"—a core feature of commonsense inference. For instance: from "Birds fly" and "Tweety is a bird," classical logic derives "Tweety flies"; but if the new information "Tweety is a penguin" is added, classical logic cannot gracefully retract the earlier conclusion, whereas non-monotonic logic can. Problem-solving power is the strongest selective pressure.); **social acceptance** (the dissemination of a symbolic system also depends on the acceptance of the academic community. Intuitionistic logic long remained marginal in mathematics not because it was technically inferior, but because it challenged the mainstream philosophical presuppositions of mathematical realism).

Conclusion: The facet of invention of logic is precisely **representational forms with symbols as their stable carriers**—the symbolic systems of propositional variables, connectives, quantifiers, inference rule formatting, truth tables, and so on. They are the "noetic mode" of the logical system, that is, the symbolic shell through which the rules are processed, transmitted, and evolved by human cognitive systems. Humans **invent** and continually modify these symbolic systems. This explains the diversity, historical evolution, and revisability of logic.

3.5 The Two Facets Are Inseparable

Above, we have separately deduced the facet of discovery and the facet of invention of logic. We must now demonstrate that these two facets are not two independent "parts" but two inseparable facets of one and the same logical system.

Argument One: The consequence of lacking the facet of discovery.

Suppose we attempt to retain only the facet of invention—that is, to regard logic merely as a set of arbitrary symbolic conventions. We immediately face an unanswerable question: Why does this set of conventions possess **normativity**?

If logical rules are merely conventions, why can we not convene a different set of rules—say, a set that permits contradictions to be true—and use it to think about the world equally effectively? The conventionalist reply is typically "because that would lead to contradictions." But why is "leading to contradictions" unacceptable? This very reply already presupposes the normativity of the law of non-contradiction—that is, it presupposes the very facet of discovery it seeks to deny.

Without the facet of discovery, logical symbols would degenerate into arbitrary game rules, forfeiting their "must-be-so" compulsory force.

Argument Two: The consequence of lacking the facet of invention.

Suppose we attempt to retain only the facet of discovery—that is, to admit only the rules themselves while denying the artificiality and variability of symbolic systems. We then cannot explain why it took humanity over two millennia to develop the symbolic system of modern predicate logic.

If rules as such were the only existents, their discovery should have been a one-time event. But the actual course of logical history shows that the symbolic representation of the same set of fundamental rules has undergone a long, trial-laden evolutionary process. Syllogistic, propositional logic, predicate logic, non-classical logics—these are different symbolic representational forms of the same set of deep

rules. Their differences cannot be explained by "discovery of the rules themselves," but only by "invention and evolution of symbolic systems."

Without the facet of invention, logic would become a static, ahistorical, mysterious entity incapable of explaining its own diversity.

Argument Three: The dual unity of rules and representation.

Synthesizing the above, the complete ontological picture of the logical system is as follows:

- **The facet of discovery (rules as such)** provides logic with its normative skeleton—it is the "bones" of logic, determining what constitutes valid inference. Non-contradiction is not a convention but a constitutive condition of any stably assertable system.
- **The facet of invention (symbolic representation)** provides logic with its operable morphology—it is the "flesh" of logic, enabling the rules to be written down, transmitted, taught, and revised. Symbolic systems are variable, optimizable, and evolvable.

Rules and representation: neither can be dispensed with. Without rules, symbols are merely meaningless ink marks; without symbols, rules cannot enter public human cognition.

This, then, is the Duality Theorem of Logic: **The logical system is the dual unity of rules as such (facet of discovery) and symbolic representation (facet of invention). The two facets are not mutually exclusive options but two inseparable facets of a single system.**

As for semantic interpretation (Level Three) and practical application (Level Four), these are the coupled products of the two facets in specific contexts. Semantic interpretation—e.g., truth conditions, model-theoretic semantics—must be anchored both in rules as such (facet of discovery: what constitutes valid truth-transmission) and in symbolic representation (facet of invention: what formal tools are used to characterize semantics). Practical application—e.g., legal argumentation, scientific reasoning—is even more clearly so: a concrete act of reasoning must both obey the fundamental rules such as non-contradiction (facet of discovery) and employ a specific symbolic system for its expression (facet of invention). The existence of Levels Three and Four further confirms the universality of duality: every "actual use" of logic is a joint presence of rules as such and symbolic representation.

3.6 Comparison with Existing Positions

Before concluding this section, it is useful to compare briefly the position of this paper with existing discovery-ism and invention-ism, in order to clarify its theoretical location.

Position	Core Claim	Phenomena Explained	Phenomena Unexplained
Discovery-ism (Realism)	Logic mirrors the structure of objective reality	Compulsory force of logic, cross-cultural universality	Diversity of logical symbols, historical evolution
Invention-ism (Conventionalism)	Logic consists of arbitrary human symbolic conventions	Diversity of logical symbols, historical evolution	Compulsory force of logic, cross-cultural universality

Position	Core Claim	Phenomena Explained	Phenomena Unexplained
This paper (Duality)	Logic is the dual unity of rules as such (facet of discovery) and symbolic representation (facet of invention)	All of the above	—

The position of this paper is not a "middle way" between discovery-ism and invention-ism, but a direct dissolution of the "either-or" framework they share. Discovery-ists have glimpsed rules as such—logic does indeed possess a normative kernel independent of symbolic convention. Invention-ists have glimpsed symbolic representation—logical symbols are indeed human creations that evolve over history. Each has glimpsed one facet of the duality of logic but mistaken that facet for the whole.

Quine's thesis of the "revisability of logic" finds a repositioning within this framework. Quine is correct in seeing that logical symbol systems can be revised under empirical pressure—this is the insight of the facet of invention. But he fails to distinguish between revisions that are adaptive adjustments of symbolic systems to specific domains (e.g., the restriction of the distributive law in quantum logic, the restriction of excluded middle in intuitionistic logic) and would-be "revisions" that would in fact entail the collapse of the assertable system itself (e.g., attempting to abolish non-contradiction). The former are healthy evolution; the latter are systemic self-destruction. The duality framework provides a criterion for making this distinction: any "revision" that does not violate non-contradiction and identity belongs to normal variation at the facet of invention; any "revision" that challenges these two bottom lines will be eliminated by evolution because it cannot form a stably assertable system.

3.7 Formalized Statement

For ease of subsequent reference and deduction, the core conclusion of this section may be formulated as a theorem:

Theorem (The Duality Theorem of Logic)

The logical system simultaneously manifests a dual existence as a facet of discovery and a facet of invention:

- **Facet of Discovery:** Rules as such—the fundamental laws of non-contradiction, identity, and causality. They are constitutive conditions for any stably assertable system, independent of symbolic representation, and endow logic with normativity.
- **Facet of Invention:** Representational forms with symbols as their stable carriers—propositional variables, connectives, quantifiers, inference rule formatting, truth tables, and other symbolic systems. They are the noetic forms through which the rules are written down, transmitted, and evolved, shaped by path-dependence and revisable according to cognitive demands.

The two facets are inseparable: without the facet of discovery, normativity is lost; without the facet of invention, operability and evolvability are lost.

4. The Facet of Discovery: Rules as Such

Section 3 has established, through axiomatic deduction, the facet of discovery of logic—namely, rules as such. The task of this section is to provide a deeper philosophical elaboration of this facet of discovery, demonstrating why the laws of non-contradiction, identity, and causality are constitutive conditions for any stably assertable system, rather than products of human convention.

Before unfolding the specific arguments, it is necessary to offer a preliminary clarification of the internal stratification within the "facet of discovery." As understood in this paper, the "facet of discovery" encompasses two kinds of rules that differ somewhat in their nature:

Type One: Constitutive rules of logic. The laws of non-contradiction and identity belong to this type. They are constitutive conditions for any stably assertable system—a system that fails to comply with them simply cannot exist as an assertable system at all. They are logical necessities.

Type Two: Universal frameworks of rational inference. The law of causality belongs to this type. It is not a constitutive rule of pure formal logic (a formal system that does not deal with causation is perfectly legitimate logically), but it is a framework upon which any substantive human reasoning about the world must necessarily rely. Its universality is not a logical necessity, but rather the product of the long-term interaction between human cognitive systems and a world possessed of causal structure. In this sense, it is "discovered" rather than "invented."

This distinction does not affect the fundamental positioning of causality within the facet of discovery, but it clarifies its special status within that facet. The following subsections will unfold in turn.

4.1 Non-Contradiction as a Condition of Existence

The law of non-contradiction is usually formulated as: "The same proposition cannot be both true and false at the same time," or formalized as $\neg(P \wedge \neg P)$. In logic instruction, it is taught as a fundamental rule of inference. But its status extends far beyond this—it is a constitutive condition that any stably assertable system must satisfy.

4.1.1 Aristotle's Foundation

Aristotle's discussion of non-contradiction in the *Metaphysics* is the classic defense of this law in the history of Western philosophy. He pointed out that non-contradiction is "the most certain of all principles," because anyone who engages in meaningful assertion already presupposes it. One who attempts to deny non-contradiction must either remain silent (thereby abandoning assertion) or open one's mouth to speak—but the moment one speaks, one's very act of assertion distinguishes between "what one has said" and "what one has not said," thereby already tacitly affirming the validity of non-contradiction.

Aristotle's argument reveals a crucial insight: non-contradiction is not merely one logical rule among many, but a constitutive condition of the very act of assertion. The denial of non-contradiction is a "performative contradiction"—the content of the assertion denies the very condition that makes the act of assertion possible.

4.1.2 As a Condition of Systemic Stability

From the perspective of systemic ontology, Aristotle's insight can be further generalized: non-contradiction is not only a condition of meaningful assertion, but a condition for any stably existing system of distinctions whatsoever.

Consider the simplest possible system of distinctions: it must be capable of distinguishing "A" from

"not-A." If this distinction is unstable—that is, if the system can simultaneously mark the same thing as both A and not-A—then the distinction itself collapses. A system that cannot maintain distinctions cannot perform any determinate information-processing function. It logically annihilates itself.

This is not the result of human "stipulation," but a constitutive constraint of systemic existence. Just as a physical measurement system, in order to function, must satisfy the stability condition that "the same quantity under the same conditions yields the same reading," so too a logical system of distinctions, in order to function, must satisfy the stability condition that "the same proposition cannot simultaneously be assigned both the value true and the value false." This condition is not an externally imposed norm, but an internal determination of the system's ability to exist as that system.

4.1.3 Dialogue with Conventionalism

A conventionalist might respond: non-contradiction is "inviolable" only because we have chosen to use it, not because it possesses any status transcending convention.

But this response overlooks a crucial point: the very act of convention already presupposes non-contradiction. When two parties convene the meaning of "P," they must be able to distinguish between "agreeing to P" and "not agreeing to P"—otherwise the act of convention cannot proceed. Any attempt to ground non-contradiction in convention inevitably falls into circularity: convention requires distinction, and distinction requires non-contradiction. Hence, non-contradiction cannot be "convened"; it is a precondition for any act of convention whatsoever.

This does not mean that non-contradiction is "a priori" or "innate." Its foundation lies not in an innate structure of the human mind (as Kant envisioned), but in the functional constraints that any stable system of distinctions—whether human cognitive systems, animal perceptual systems, or artificial information-processing systems—must satisfy. In this sense, non-contradiction is discovered: in the long course of cognitive practice and reflection, humans have progressively made this rule, immanent in every act of assertion and distinction, explicit, endowing it with clear symbolic form.

4.2 Identity as a Presupposition of Assertion

The law of identity is usually formulated as $A = A$, or "Everything is identical with itself." In logical calculi, it appears as a trivial tautology. Yet the philosophical significance of identity lies precisely in its apparent "triviality"—it is so fundamental that every act of assertion already tacitly presupposes it.

4.2.1 Identity and the Stability of Symbols

Consider the simplest scenario of symbol use: we use the symbol "S" to refer to an object O. If the meaning of "S" could drift during a process of reasoning—that is, if "S" in an earlier step referred to O, while "S" in a later step referred to a different object O'—then the inferential chain would break. The conclusion drawn from the premises would no longer be a conclusion about O, but about some indeterminate object.

What the law of identity guarantees is precisely the stability of symbols during reasoning. It demands that, within a single inferential chain, the same symbol must maintain the same reference, and the same proposition must maintain the same truth-value. Without this guarantee, no reasoning extending beyond a single step could proceed—because after each step, the meaning of the symbols might already have altered.

4.2.2 Dialogue with Quine

Quine is famous for his critique of the analytic/synthetic distinction and his thesis of the revisability of logic. He suggests that, in extreme cases, we might even revise logical laws, including the law of identity.

However, the force of Quine's "revisability" thesis lies in the fact that it can be understood as a thesis about **symbolic systems**, rather than about **rules as such**. When we "revise identity," we are in fact doing one of two different things: either we are replacing an old symbolic system with a new one (this belongs to evolution at the facet of invention); or we are arguing, at a meta-level, that "identity should be revised"—and this very argument presupposes the validity of identity at the meta-level.

Quine is correct in seeing that logical symbolic systems are revisable—this is the insight of the facet of invention. But he fails to distinguish between the variability of symbolic systems and the constitutive status of rules as such. The status of identity as "a presupposition that makes assertion possible" is not shaken by altering the symbols we use to write it down.

4.3 Causality: The Facet of Discovery of Rational Inference

Thus far, we have argued that non-contradiction and identity are constitutive conditions for any stably assertable system—they are logical necessities. The status of causality, treated in this subsection, differs: it is not a constitutive law of pure formal logic, but it is an indispensable component of the universal framework of human rational inference.

Two questions must be clearly distinguished:

- **Must a logical system include causality?** No. Pure formal logic (e.g., propositional calculus, predicate calculus) does not deal with causation; its validity is independent of the law of causality.
- **Must human rational inference rely on causal structure?** Yes. Any substantive reasoning about the world that goes beyond purely formal calculation necessarily involves causation—from everyday decisions ("If I press this switch, the light will come on") to scientific explanations ("Smoking causes lung cancer"), all presuppose the validity of causal inference.

In this sense, causality is not "the facet of discovery of logic" (that belongs to the domains of non-contradiction and identity), but rather "the facet of discovery of rational inference"—it is an inferential framework that humans have discovered and internalized through long-term interaction with a world possessed of causal structure. The normative source of this framework is precisely what this subsection aims to demonstrate.

4.3.1 Hume's Insight and Its Limits

Hume correctly saw that the "necessity" of causality cannot be directly observed in any single experience. No matter how many times we see the sun rise or hear the alarm clock ring, what we observe is only temporal succession, not logical necessity.

But Hume's conclusion—that causality is merely psychological habit—leaves an unanswerable question: Why is this psychological habit so universal? Why do all human cultures, and even certain animals, exhibit the capacity for causal reasoning? If causality were merely arbitrary psychological association, why does it succeed in guiding our actions in the world?

4.3.2 Causality as the Mirroring of World-Structure

The answer provided by systemic ontology is this: causality is the structural adaptation of human

cognitive systems to the self-organizing tendency of the world itself.

The world is not a completely chaotic Brownian motion. It exhibits stable causal structure—certain types of events regularly lead to certain types of outcomes. This is not a logical necessity (like non-contradiction), but a factual regularity. Yet this regularity is not arbitrary—it is a manifestation of the self-organizing tendency of the cosmos: matter condenses under gravity into stars, stars produce heavy elements through nuclear fusion, heavy elements constitute planets and life... This is a directed chain that can be traced causally.

The human capacity for causal reasoning is a cognitive tool that evolved through long-term interaction with a world possessed of such causal structure. It is neither a mysterious intuition of "necessary connection" (as rationalists imagined) nor an arbitrary psychological association (as Hume believed), but the statistical tracking and structured representation of the causal density actually operative in the world.

In this sense, causality is "discovered": over hundreds of thousands of years of practical existence, humans have progressively internalized the causal structure of the world into the inferential rules of their cognitive frameworks. It is not a logical necessity, but it is a fact about the world—and the stability of this fact suffices to underwrite the normativity of causality as a reliable reasoning tool.

4.4 Summary: The Unified Picture of the Facet of Discovery

To summarize this section, the facet of discovery of logic—rules as such—comprises three levels of existential conditions:

Law	Nature	Status	Stratum
Non-Contradiction	Constitutive condition for any stable system of distinctions	Logical necessity	Constitutive rules of logic
Identity	Presupposition for referential stability in any symbolic system	Logical necessity	Constitutive rules of logic
Causality	Structural mirroring of the self-organizing tendency of the world; universal framework of human rational inference	Factual regularity (not logically necessary, but possessing cross-cultural universal normativity)	Facet of discovery of rational inference

This stratification does not affect the "discovery" status of causality—like non-contradiction and identity, it is a normative framework discovered by humans in cognitive practice, independent of symbolic convention, rather than an arbitrarily invented set of symbolic game rules. But it clarifies the scope of applicability of causality: it is not a constitutive law of pure formal logic, but a universal framework of substantive human reasoning.

5. The Facet of Invention: Representational Forms with Symbols as

Their Stable Carriers

Section 3 has established the facet of invention of logic—namely, representational forms with symbols as their stable carriers. The task of this section is to provide a deeper historical and mechanistic elaboration of this facet of invention, demonstrating why logical symbolic systems are human inventions and how they undergo variation, selection, and evolution within the noetic ecology.

5.1 One Rule, Multiple Symbols

The most immediate lesson from the history of logic is this: the same set of fundamental logical rules can be represented by radically different symbolic systems.

5.1.1 Three Symbolic Expressions of Non-Contradiction

Consider the law of non-contradiction as an example. In Aristotle's *Metaphysics*, it is formulated as:

"The same attribute cannot at the same time belong and not belong to the same subject in the same respect."

This formulation employs natural language and depends upon the subject-predicate grammatical structure of ancient Greek. Its advantage is intuitive accessibility; its disadvantage is lack of precision and difficulty of formal manipulation.

In medieval scholastic philosophy, non-contradiction was incorporated into the framework of term logic and formulated as:

"No proposition is true at the same time as its contradictory."

This formulation shifts non-contradiction from "attributes of things" to "truth-values of propositions," marking a turn from ontology toward linguistic analysis. Yet it remains constrained by the grammatical forms of natural language.

In modern propositional logic, non-contradiction is formalized as:

$$\neg(P \wedge \neg P)$$

This symbolic expression is thoroughly liberated from the shackles of natural grammar. P can stand for any proposition; $\neg$ and $\wedge$ are precisely defined truth-functional operators. This symbolization enables non-contradiction to be incorporated into a fully formalized calculative system, becoming a mechanically checkable rule of inference.

5.1.2 The Lesson of Symbolic Difference

All three expressions point to one and the same rule. The difference lies not in the rule itself but in its representational forms.

This fact reveals a core feature of the facet of invention: **the independent variability of symbolic systems**. The rule itself (facet of discovery) is invariant—the normative content of non-contradiction as a constitutive condition for any stably assertable system does not change whether we write it in Greek, Latin, or mathematical symbols. But symbolic representation (facet of invention) is variable—it can be optimized, replaced, and adapted to different cognitive demands and technological media.

If logic were purely "discovered," the symbolic systems of all civilizations should tend toward convergence, just as all counting systems eventually adopted positional notation. But the actual course of logical history shows that the evolution of symbolic systems is not simple convergence, but exhibits rich diversity and path-dependence.

5.2 The Artificiality and Path-Dependence of Symbolic Systems

The concrete morphology of logical symbolic systems is shaped by multiple human factors and historical contingencies. This is not a "defect" but a natural feature of the evolution of noetic genes.

5.2.1 The Shaping Force of Linguistic Background

The subject-predicate structure of Aristotelian syllogistic has a deep affinity with the grammatical structure of ancient Greek. In Greek, a typical declarative sentence consists of a subject (*ὑποκείμενον*, "that which lies beneath") and a predicate (*κατηγορούμενον*, "that which is said of the subject"). Aristotle elevated this grammatical structure into a logical category: all propositions can be reduced to the form "S is P," and all reasoning can be reduced to the syllogistic form.

This framework was so successful that for two millennia, logic was nearly synonymous with syllogistic. Yet it also imposed profound limitations: relational propositions (e.g., "A is greater than B") are difficult to handle within a subject-predicate structure; the core tools of modern logic—multiple predication, nested quantifiers—cannot be naturally expressed within the syllogistic framework.

It was not until Frege, in his 1879 *Begriffsschrift*, replaced the subject-predicate structure with a function-argument structure ($f(x)$) that logic was liberated from the grammatical shackles of natural language. Frege's innovation was not a revision of "rules as such"—non-contradiction remained just as valid after him as before—but a fundamental reconstruction of the symbolic representational system.

5.2.2 The Filtering Effect of Technological Media

The evolution of logical symbolism has also been profoundly influenced by the media of writing and transmission. Frege's *Begriffsschrift* employed a two-dimensional diagrammatic structure, using horizontal and vertical lines to represent logical relations. This design was conceptually elegant but technologically unsuccessful—the printing technology of the time could not economically reproduce such complex two-dimensional diagrams.

In contrast, the linear symbol system adopted by Peano and Russell (e.g., $\supset$ for implication, $\sim$ for negation), though less intuitively perspicuous than Frege's diagrams, was easy to typeset, print, and disseminate. The selective pressure of the technological medium ultimately made the linear symbol system the standard format of twentieth-century logic.

This case reveals an important principle: **the "victory" of a symbolic system is not purely a matter of logical superiority, but the combined result of logical efficacy, technological feasibility, and ease of dissemination.** This is a characteristic feature of noetic gene selection—selective pressures come from multiple dimensions, not a single criterion of truth.

5.2.3 The Driving Force of Cognitive Demands

The evolution of logical symbolism is also driven by the cognitive demands of different eras. Aristotelian logic served the practices of debate and classification in ancient Greece—syllogistic provided a tool for testing the validity of arguments. Medieval scholastic logic served theological argumentation—the subtle distinctions of term logic (e.g., the distinction between "meaning" and "reference") arose from the need for rigorous analysis of theological propositions.

The birth of modern predicate logic was directly driven by the pursuit of rigor in the foundations of mathematics. Frege's goal was to provide a purely logical foundation for arithmetic, which required a symbolic system far more powerful than syllogistic—capable of expressing multiple relations, nested

quantifiers, and higher-order predicates. His *Begriffsschrift* was designed precisely for this purpose. The non-classical logics that emerged in the latter half of the twentieth century are even more striking examples of cognitive demands driving symbolic evolution:

- **Fuzzy logic:** Invented to handle vague concepts like "tall" or "middle-aged," replacing the two-valued $\{0,1\}$ with the continuous interval $[0,1]$.
- **Intuitionistic logic:** Arising from Brouwer's philosophical reflections on mathematical constructivity, rejecting the unconditional applicability of excluded middle over infinite domains.
- **Quantum logic:** Proposed to handle non-classical phenomena in quantum mechanics (such as superposition and measurement), modifying the distributive law of classical logic.
- **Non-monotonic logic:** Invented to handle commonsense reasoning in artificial intelligence, allowing conclusions to be retracted as new information is added.

These "logical variations" are not denials of rules as such—non-contradiction remains in force in some form within fuzzy logic and intuitionistic logic. They are adaptive adjustments at the level of symbolic representation: designing reasoning tools better suited for specific cognitive domains.

5.2.4 The Status of Excluded Middle: A Test Case for the Duality Framework

Quine's thesis of the "revisability of logic" finds its most famous challenges in the questioning of excluded middle by quantum mechanics and in Brouwer's intuitionistic rejection of excluded middle. This provides an ideal test case for the duality framework: Does excluded middle belong to the facet of discovery (rules as such) or to the facet of invention (symbolic representation)?

The answer offered in this paper is that it depends on the level at which excluded middle is understood.

Excluded middle as symbolic representation: In classical propositional logic, excluded middle is formalized as $P \vee \neg P$ and assigned the semantic interpretation that "for any proposition P , either P or $\neg P$ must be true." This symbolic expression and its semantic interpretation belong to the facet of invention—they constitute a set of symbolic conventions invented by humans for handling deterministic reasoning. When Brouwer rejects the unconditional applicability of excluded middle over infinite domains, he is in fact challenging the scope of applicability of this symbolic convention, not denying any validity of excluded middle whatsoever. Intuitionistic logic replaces the symbolic conventions of classical logic with a different symbolic system (e.g., negation is redefined) and a different semantic interpretation ("truth" is understood as "constructive provability"). This is variation and selection at the level of the facet of invention—one symbolic system is replaced by another because it fails to meet the cognitive demands of a particular domain.

Excluded middle as rule as such? Does there exist a "rule of excluded middle as such" that transcends symbolic representation? This is a more complex question. This paper inclines toward the view that excluded middle does not, like non-contradiction and identity, constitute an absolute condition for any stably assertable system. A mathematical system that adopts intuitionistic logic remains a stably assertable system—it merely refuses to apply excluded middle unconditionally over infinite domains, while still observing some form of excluded middle over finite domains (because of decidability over finite domains). In this sense, excluded middle is more akin to a **principle of cognitive efficiency**—in deterministic domains, it is an efficient reasoning tool; in non-deterministic domains, it requires revision or restriction. Its universality is less than that of non-contradiction, yet it is not an arbitrary convention—it is anchored in the deterministic structures that the world exhibits at certain scales.

This analysis demonstrates that Quine's "revisability of logic" is entirely correct at the level of the facet

of invention—logical symbolic systems are indeed revisable under empirical pressure. But Quine's error lies in extending this revisability to all logical rules, including those that constitute absolute conditions for stably assertable systems (such as non-contradiction). The duality framework permits a more refined distinction: the facet of invention is revisable; the facet of discovery (at least its core) is not revisable—not because we "forbid" revision, but because any "logical system" that attempts to revise these core rules will be unable to operate as a stably assertable system.

5.2.5 The Imprint of Historical Contingency

The establishment of certain symbolic conventions is due not to absolute "superiority" but to the choices of key figures, the authority of particular texts, or contingencies of transmission pathways.

The now-universal symbols for logical connectives— $\neg$ (negation), $\wedge$ (conjunction), $\vee$ (disjunction), $\rightarrow$ (implication)—were progressively established in the early twentieth century by Peano, Russell, Hilbert, and others. Before this, Frege had used a completely different set of symbols (e.g., $\vdash$ for assertion, $\supset$ for conditional). The victory of the Peano-Russell system was due partly to its intrinsic merits (conciseness, ease of typesetting), partly to the enormous influence of Russell and Whitehead's *Principia Mathematica*, and partly to the promotion efforts of the Hilbert school.

Had history been slightly different—had Frege's *Begriffsschrift* received more attention during his lifetime, or had Peano's system not been taken up by Russell—the symbols in today's logic textbooks might look quite different. This is not to say that the evolution of symbolic systems is entirely arbitrary, but rather that, among multiple functionally equivalent symbolic schemes, the one that ultimately prevails often bears the imprint of historical contingency.

5.3 Symbolic Systems as Noetic Genes

From an evolutionary perspective, logical symbolic systems are quintessential "**noetic genes**" —units of symbolic information capable of replication, variation, and selection within cognitive systems. This perspective provides a unified analytical framework for understanding the history of logic.

5.3.1 Replication: The Intergenerational Transmission of Symbols

Logical symbols are transmitted across generations and across cultures through education, texts, and technological media. Every student who learns logic replicates these symbolic patterns and their associated inference rules within their own cognitive system.

This replication process is not a simple "copying." In learning $\neg(P \wedge \neg P)$, the student not only memorizes the written form of the symbols but also internalizes the operational rules associated with them (how to read a truth table, how to perform negation and conjunction). This "operational internalization" is a key difference between the replication of noetic genes and biological genes: biological genes transmit only information; noetic genes transmit, along with information, the capacity for information processing.

5.3.2 Variation: The Creative Innovation of Symbols

Logical symbols undergo variation in transmission. Sources of variation include:

- **Creative innovation:** Frege's invention of quantifier symbols was a fundamental symbolic variation. He replaced the cumbersome natural-language "for all x " and "there exists an x " with $\forall x$ and $\exists x$, thereby enabling nested quantifiers and multiple predication.
- **Cross-domain transfer:** Boole transferred algebraic symbolism into the domain of logic,

creating Boolean algebra. He used $\times$ for conjunction and $+$ for disjunction—a symbolic choice rooted in his analogy with algebra rather than any intrinsic requirement of logic.

- **Adaptive adjustment:** Zadeh, in order to handle vague concepts, extended the truth-value set of classical logic from $\{0,1\}$ to the interval $[0,1]$ and redefined the operational rules of the connectives (e.g., using min for conjunction). This was a key variation enabling the symbolic system to adapt to a new cognitive domain.
- **Misreading and reinterpretation:** In the history of logic, "creative misreadings" of predecessors by successors are often a source of variation. Brouwer's rejection of excluded middle can be seen as a "misreading" of the rules of classical logic—proceeding from his philosophical stance of mathematical constructivity, he discerned the limitations of classical logic's applicability over infinite domains.

5.3.3 Selection: The Filtering of the Cognitive Ecology

Not all symbolic variations survive. Selective pressures come from multiple dimensions:

- **Cognitive efficiency:** Can the symbolic system express complex reasoning concisely? A mathematical proposition that modern predicate logic can express in a few short lines might require pages of cumbersome derivation within the syllogistic framework. This efficiency advantage was a key factor in predicate logic's selective advantage.
- **Systemic compatibility:** Can the new symbolic system integrate with existing bodies of knowledge? Fuzzy logic has been widely accepted partly because it is a "conservative extension" of classical logic—when membership degrees are restricted to $\{0,1\}$, fuzzy logic collapses back into classical logic. This compatibility reduces the acceptance cost for the academic community.
- **Problem-solving power:** Can the new symbolic system solve problems that the old system could not effectively handle? Non-monotonic logic has gained traction in AI precisely because classical logic cannot handle "defeasible reasoning"—a core feature of commonsense inference. (For example: from "Birds fly" and "Tweety is a bird," classical logic derives "Tweety flies"; but if the new information "Tweety is a penguin" is added, classical logic cannot gracefully retract the earlier conclusion, whereas non-monotonic logic can. Problem-solving power is the strongest selective pressure.)
- **Social acceptance:** The dissemination of a symbolic system also depends on the acceptance of the academic community. Intuitionistic logic long remained marginal in mathematics not because it was technically inferior, but because it challenged the mainstream philosophical presuppositions of mathematical realism.

5.3.4 The Evolutionary Picture: Non-Teleological Progress

Synthesizing the above, the evolution of logical symbolic systems presents a picture of **non-teleological progress**:

- It **is** progressive—in expressiveness, conciseness, and operability, modern predicate logic is indeed superior to Aristotelian syllogistic. This is not cultural-relativist "stylistic change," but objectively assessable improvement in cognitive efficiency.
- But it is **not** teleological—there is no preordained "correct path" guiding logical symbols from syllogistic necessarily toward predicate logic. The evolutionary path is replete with historical contingencies, path-dependence, and unpredictable variations. Frege's function-argument

structure was not the "necessary development" of syllogistic, but a fundamental framework reconstruction whose emergence depended on Frege's individual genius, the historical occasion of the crisis in the foundations of mathematics, and the filtering effect of printing technology on symbolic forms.

This picture perfectly embodies the core insight of the facet of invention: logical symbolic systems are noetic genes created by humans, and their evolution follows the ecological dynamics of variation and selection, rather than a unidirectional process of "discovering truth."

5.4 Dialogue with Conventionalism

Before concluding this section, it is useful to compare briefly the position of the facet of invention with classical conventionalism, in order to clarify its theoretical location.

The core claim of conventionalism (e.g., Carnap) is that logical rules are artificially convenable. Which logical system we choose depends on convenience, not truth. Logic is not a matter of "right or wrong," but of "useful or not."

This paper's amendment is as follows: Conventionalism correctly discerns the artificiality and selectability of symbolic systems, but it erroneously extends this insight to rules as such. The distinction drawn in this paper is:

- **Rules as such (facet of discovery) are not convenable.** Non-contradiction is valid not because we have convened it, but because it is a constitutive condition for any stably assertable system. We cannot abolish non-contradiction by "re-convening" it, any more than we can alter universal gravitation by re-convening it.
- **Symbolic representation (facet of invention) is convenable.** Whether we use $\neg$ or $\sim$ for negation, whether we use truth tables or proof trees—these are the results of human choice.

Different conventional systems can be functionally equivalent or can have differing merits.

Conventionalism conflates these two levels. When Carnap says "we are free to choose logical rules," he is in fact saying "we are free to choose symbolic representational systems." But this freedom of choice is constrained by rules as such—we can only choose among symbolic systems that are capable of effectively representing the meta-rules. A "logical symbolic system" that permits contradictions to be true is not a forbidden convention, but a pseudo-system that simply cannot perform the function of logic.

In this sense, the facet of invention is not "arbitrary" but constrained by the facet of discovery. Symbolic systems can vary endlessly, but they must be capable of representing the same set of fundamental rules—just as map projections can vary endlessly, but they must be capable of correctly representing the geometric structure of the Earth.

5.5 Summary: The Evolutionary Picture of the Facet of Invention

To summarize this section, the facet of invention of logic—representational forms with symbols as their stable carriers—presents the following evolutionary picture:

Feature	Description	Example
Independent variability	A single rule admits multiple symbolic representations	Non-contradiction expressed in syllogistic, propositional, and predicate forms

Feature	Description	Example
Artificial shaping	Shaped by language, technology, cognitive demands, and historical contingency	The printing failure of Frege's <i>Begriffsschrift</i>
Noetic gene	Units of symbolic information capable of replication, variation, and selection	The transfer of Boolean algebra from algebra to logic
Non-teleological progress	Directional evolution without a preset endpoint	Efficiency gains from syllogistic to predicate logic

The facet of invention endows the logical system with expressive flexibility and evolutionary adaptability. Without this facet, logic would be a dead, ahistorical, closed system incapable of self-optimization. It is precisely the existence of the facet of invention that makes logic a living part of the noetic ecology—continually reshaping its symbolic morphology in response to the shifting cognitive demands of humanity, while remaining ever anchored to the normative foundation established by the facet of discovery.

6. Corollaries and Applications

The preceding sections have established the Duality Theorem of Logic and elaborated the concrete content of the facet of discovery and the facet of invention respectively. The task of this section is to demonstrate that this framework is not merely a philosophical unification but possesses genuine deductive power—capable of generating testable empirical predictions and actionable engineering principles. We shall advance two corollaries: the first pertains to the evolutionary dynamics of logical systems in history; the second pertains to principles of logical design in the field of artificial intelligence.

6.1 Corollary One: The Ecological Dynamics of Logical Evolution

6.1.1 From Duality to Evolutionary Dynamics

If logic possesses duality—the facet of discovery being rules as such, the facet of invention being symbolic representation—how should we understand the historical transformations of logical systems? Discovery-ism cannot explain transformation. If logic were merely the discovery of rules as such, the history of logic should be a one-time event of "seeing" followed by fixation. Yet the fact is that logical symbolic systems have undergone over two millennia of continuous evolution and continue to generate new variations to this day (e.g., non-classical systems such as linear logic and separation logic).

Invention-ism cannot explain directionality. If logical symbols were merely arbitrary conventions, the transformations of logical history should have no direction—any convention would be equally valid as any other. Yet the fact is that modern predicate logic is genuinely superior to syllogistic in expressiveness and inferential efficiency, and this judgment is not cultural bias but an objectively assessable improvement in cognitive efficiency.

The duality framework provides a unified explanation: **Logical evolution is a process of directed**

variation of symbolic systems (facet of invention), constrained by rules as such (facet of discovery) and driven by selective pressures within the noetic ecology.

6.1.2 Formalized Description of the Evolutionary Mechanism

This dynamics can be described using the following framework:

Sources of Variation: Variations in logical symbolic systems arise from multiple sources—

- Creative innovation (e.g., Frege's invention of quantifier symbols)
- Cross-domain transfer (e.g., the transfer of Boolean algebra into logic)
- Adaptive adjustment to emerging cognitive demands (e.g., fuzzy logic's response to vague concepts)
- Creative misreading and reinterpretation (e.g., intuitionism's philosophical reflection on excluded middle)

Selective Pressures: Not all variations survive. Selective pressures come from—

- **Representational efficiency:** Can the new symbolic system represent the same set of rules more concisely and precisely?
- **Problem-solving power:** Can the new symbolic system solve problem types that the old system could not effectively handle?
- **Systemic compatibility:** Can the new symbolic system integrate with existing knowledge networks?
- **Social acceptance:** Can the new symbolic system be accepted and disseminated by the academic community?

Constraint Conditions: All variations must satisfy the normative constraints of the facet of discovery. No matter how novel a symbolic system may be, if it cannot effectively represent the fundamental rules of non-contradiction and identity—that is, if it cannot operate as a stably assertable system—it will not be selected. This is the inviolable boundary of evolution at the facet of invention.

Evolutionary Direction: Within the constraint conditions, selective pressures drive symbolic systems toward higher cognitive efficiency. This is not teleology—evolution has no preset "ultimate correct" endpoint—but it is directional: certain variations genuinely satisfy selective pressures better than others and are therefore more likely to survive and spread.

6.1.3 Testable Empirical Predictions

This dynamical framework yields several testable predictions:

Prediction One: When a new cognitive domain emerges that exerts pressure on existing logical symbolic systems in the form of "expressive difficulty" or "inferential inefficiency," the probability of symbolic variation in logic will rise significantly. This explains why the emergence of non-classical logics concentrated in the latter half of the twentieth century—quantum mechanics, computer science, and artificial intelligence posed new demands that classical logic could not readily meet.

Prediction Two: Among functionally equivalent symbolic schemes, the scheme that ultimately gains selective advantage will be the one that achieves an optimal balance between cognitive efficiency (conciseness, operability) and social acceptance (compatibility with existing academic networks). This explains the failure of Frege's two-dimensional diagrams and the success of the Peano-Russell linear symbols.

Prediction Three: Any "radical logical innovation" that attempts to completely discard the constraints of the facet of discovery (e.g., attempting to construct an inferential system that permits contradictions

as a mainstream logic) will fail to gain widespread acceptance—not because it is "forbidden," but because it cannot effectively perform the normative function of logic.

These predictions provide an operational analytical framework for future research in the history of logic: researchers can identify the sources of variation, selective pressures, and constraint conditions for specific events of logical symbolic change (e.g., the transition from syllogistic to predicate logic, the rise of a particular non-classical logic), thereby testing the explanatory power of this framework.

6.2 Corollary Two: The Duality Principle for AI Logic Design

6.2.1 The Logical Predicament of Contemporary AI

Current artificial intelligence systems, particularly large language models (LLMs), exhibit a conspicuous predicament with respect to logical reasoning.

On the one hand, these systems display astonishing capacities for pattern matching and language generation. They can perform analogical reasoning and commonsense inference in natural language, and even achieve high scores on certain standardized logic tests.

On the other hand, they reveal profound logical vulnerabilities: they generate logical contradictions without awareness ("hallucinations"); they cannot perform rigorous multi-step deductive reasoning; they give inconsistent answers to the same question under different formulations. These failures are not accidental "bugs" but structural consequences of the system architecture.

From the diagnostic perspective of the duality framework, the fundamental problem with current large language models is this: **they possess only the facet of invention, lacking the facet of discovery.**

Specifically:

- Large models learn from vast corpora the statistical patterns of symbol usage—which symbol sequences appear with high frequency in which contexts. This is a form of the facet of invention: they have mastered the statistical distribution of symbolic representation.
- But they lack the anchoring of the facet of discovery—the laws of non-contradiction, identity, and causality are not built into the system as "hard constraints that any inference must satisfy," but are merely "softly" learned as one kind of statistical pattern among many.

The consequence is that the model can, most of the time, generate inferences that appear reasonable, but it has no internal mechanism to ensure that it does not produce logical contradictions. It lacks a "logical spine"—an inviolable normative skeleton.

6.2.2 The Duality Design Principles

Based on the Duality Theorem of Logic, we propose the following duality principles for AI logic module design:

Principle One: Hardwiring the Facet of Discovery—An Immutable Kernel of Rules

The fundamental laws of logic—non-contradiction, identity, and the framework of causal reasoning—should be built into the logic module of the AI system as hard constraints. These constraints should not be parameters revisable by the learning process, but should constitute the "constitutional layer" of reasoning:

- Any inferential output must pass a non-contradiction check (cannot simultaneously assert P and $\neg P$)
- Any use of symbols must comply with the identity constraint (the same symbol must maintain stable reference within the same inferential chain)
- Any causal inference must satisfy a causal weight threshold (correlation does not equal

causation)

This kernel can be implemented at the hardware level (e.g., dedicated verification chips) or at the level of low-level code (an immutable core module).

Principle Two: Opening the Facet of Invention—An Evolvable Symbolic Interface

Above the hard constraints of the facet of discovery, the system should possess a learnable, evolvable symbolic system:

- Capable of loading different logical symbol systems (classical logic, fuzzy logic, non-monotonic logic, etc.) according to domain-specific demands
- Capable of learning new conceptual relations from data, but such learning results must pass verification by the facet of discovery before entering the inferential chain
- Capable of dynamically adjusting symbolic conventions to optimize communication efficiency in interaction with humans or other systems

Principle Three: Separating Verification from Inference—An Independent Axiom Audit Layer

To realize the above two principles, the system's inference module and verification module should be separated. The inference module (the "senate network") is responsible for generating inferential paths and may flexibly invoke the symbolic systems of the facet of invention; the verification module (the "framework auditor layer") is independent of the inference module and performs consistency audits on inferential outputs based solely on the hard constraints of the facet of discovery. The verification module possesses veto power—any inference that violates the facet of discovery should be rejected, regardless of its statistical probability.

This separation ensures that the system's normativity is governed not by statistics but by rules as such.

6.2.3 Comparison with Existing Approaches

Dimension	Purely Statistical Approach (LLMs)	Purely Symbolic Approach (GOF AI)	Duality Approach
Source of rules	Learned from statistical data	Hand-coded by humans	Hard constraints of facet of discovery + learnable facet of invention
Normativity	Weak (prone to contradiction)	Strong (but rigid)	Strong (hard constraints) yet flexible (evolvable)
Adaptability	Strong (learns from data)	Weak (relies on manual updating)	Strong (facet of invention is evolvable)
Interpretability	Low (black box)	High (symbolic transparency)	High (rule-anchored + symbolic transparency)
Evolutionary capacity	Requires retraining	Requires manual modification	Online evolution (facet of invention variation + facet of discovery constraint)

The problem with the purely symbolic approach (e.g., early expert systems, GOF AI) is that the symbolic system is rigid and cannot adapt to new domains. The problem with the purely statistical

approach (current LLMs) is that it lacks a normative foundation and cannot guarantee logical consistency. The duality approach attempts to combine the advantages of both: the facet of discovery provides the normative skeleton; the facet of invention provides expressive flexibility.

6.2.4 Path to Engineering Verification

These design principles are not idle speculation. In an independent technical white paper, *The Parliament of Logical Subuniverses: An AGI Alignment Architecture Based on Systemic Dynamics* (He, 2026), the above principles have been concretized into a complete architectural design:

- **Axiom System (Facet of Discovery):** Six root axioms (including logical axioms such as causal consistency, conceptual consistency, and situational matching) are built in as hardware-immutable foundations.
- **Dynamic Shared Symbolic Concept Base (Facet of Invention):** Symbolic concepts and their relations can evolve dynamically according to task demands, but all updates must pass axiom verification.
- **Framework Auditor AGI (Separation of Verification):** Independent of the inference module, it monitors all inferential outputs in real time and possesses veto power.

This architecture provides a proof of concept for the engineering feasibility of the duality design principles. Interested readers are referred to that white paper for complete technical details; they are omitted here to avoid distracting from the philosophical focus of this paper.

6.3 Summary

This section has derived two corollaries from the Duality Theorem of Logic:

- **Corollary One (Ecological Dynamics of Logical Evolution):** It interprets the history of logic as a process of directed variation of symbolic systems (facet of invention), constrained by rules as such (facet of discovery) and driven by selective pressures. This framework explains phenomena that neither discovery-ism nor invention-ism can account for alone, and yields testable empirical predictions.
- **Corollary Two (Duality Principle for AI Logic Design):** Based on the Duality Theorem, it proposes that AI logic modules should possess both hard constraints of the facet of discovery (an immutable kernel of rules) and evolvability of the facet of invention (a learnable symbolic interface), with verification and inference separated. This principle offers a new design paradigm for overcoming the logical predicament of current AI.

Together, the two corollaries demonstrate that the Duality Theorem of Logic is not only a philosophical unification but also possesses genuine deductive power—it can generate empirical predictions about logical evolution and yield actionable engineering principles.

7. Meta-Reflection: The Duality of This Argument and Its Loop of Legitimation

Any meta-theoretical discourse on the nature of logic confronts an inescapable problem of self-reference. This paper is no exception. The task of this section is to face this problem squarely and to demonstrate how this paper achieves self-legitimation without falling into paradox. This not only

safeguards the integrity of the argument but also constitutes the highest-order exemplification of the Duality Theorem of Logic.

7.1 Self-Reference: The Touchstone of Meta-Theory

The core claim of this paper is that the logical system exhibits duality—the facet of discovery being rules as such, the facet of invention being symbolic representation. Let us now apply this claim to this paper itself:

- **What is the facet of discovery of this paper?** It is the logical rules that this paper itself follows—the laws of non-contradiction, identity, and the validity of causal inference. If the argument of this paper violated these rules, it would be nonsense and could not be accepted by any rational reader.
- **What is the facet of invention of this paper?** It is the symbolic system employed to express the argument—Chinese natural language, logical symbols (such as $\neg$, $\wedge$, $\vee$, $\rightarrow$), section structure, terminological choices, and case selections. These are the symbolic organizational forms that the author has "invented" to convey the argument.

The argument of this paper is itself an instance of a logical system—it uses logical rules (facet of discovery) and symbolic representation (facet of invention) to speak about "what logic is." Hence, this paper inevitably confronts a problem of self-reference: **Whence comes the legitimacy of this very argument concerning the duality of logic?**

If the argument of this paper is valid, then the law of duality that it reveals should also apply to itself. But does this mean that this paper falls into circular argumentation—using logic to prove a conclusion about logic and then applying the conclusion back to the argument itself?

This is not a question that can be evaded. Any meta-theoretical discourse that cannot account for the source of its own legitimacy leaves a logical gap: it demands that the reader accept a standard that it has not itself satisfied.

A survey of existing meta-theoretical discourses reveals roughly four strategies for handling this issue, each with its own predicament:

- **Hierarchical Evasion:** Claiming that "this paper is meta-language and its object is object-language, so there is no self-reference." The problem with this strategy is that it merely pushes the question up one level—whence comes the legitimacy of the meta-language? Tarski's hierarchy can retreat indefinitely without ever founding the legitimacy of the bottom level.
- **Pragmatic Bracketing:** Declaring that "the problem of self-reference lies beyond the scope of this paper and will not be addressed." This concedes the incompleteness of the argument. A reviewer or attentive reader may legitimately ask: if your theory cannot account for itself, why should we accept it?
- **Rhetorical Obscurity:** Using literary language to hint that self-reference has been handled without providing rigorous argumentation. This strategy cannot withstand scrutiny; once exposed, the paper's rigor is called into question.
- **Relativist Dissolution:** Claiming that "all theories are relative, and this paper is no exception." The problem with this strategy is that it undermines itself—if the conclusion of this paper is only relative, then on what grounds does it demand that the reader accept it rather than the opposite view? The normative force of the theory is thereby forfeited.

The aim of this paper is not evasion, but to achieve a loop of self-reference squarely—to acknowledge

the existence of self-reference while indicating a path that does not fall into circularity or paradox.

7.2 The Self-Reference Loop of This Argument

The solution offered in this paper proceeds in three steps: distinguishing two forms of self-reference; relocating the criterion of legitimacy from within the system to outside it; and demonstrating this paper's own self-consistency as an instance of duality.

7.2.1 Distinguishing Two Forms of Self-Reference

First, it is necessary to distinguish two forms of self-reference that are fundamentally different in nature.

Intra-systemic self-reference refers to the construction, within a formal system and by means of encoding, of a proposition that refers to itself. Gödel's incompleteness theorems are based on precisely this kind of self-reference: he constructed a proposition G such that G asserts " G is unprovable within the system." This kind of self-reference leads to paradox or undecidability because it attempts to achieve adjudication of the system within the system itself—it is like having a judge preside over his own case.

Meta-level self-reference refers to speaking about an object-system as a whole from a higher linguistic level. When we write a paper analyzing "the nature of logic," we are using a meta-language (Chinese natural language plus logical symbols) to speak about an object-language (the logical system). This kind of self-reference does not necessarily lead to paradox, because we, as speakers, stand outside the object-system.

The argument of this paper belongs to meta-level self-reference, not intra-systemic self-reference. This paper does not attempt to construct a self-referential proposition within the logical system, but stands outside the logical system and employs a set of meta-linguistic tools to analyze the overall structure of the logical system. Hence, this paper does not necessarily fall into a Gödelian paradox.

But distinguishing the two forms of self-reference is only the first step. It explains why this paper is not "automatically paradoxical," but it does not explain whence the legitimacy of this paper comes. Meta-level discourse still requires a criterion of legitimacy—otherwise anyone could claim that their own meta-theory is "correct."

7.2.2 External Adjudication: Relocating Legitimacy from Within the System to Outside It

The solution proposed in this paper is this: **the legitimacy of a meta-theoretical argument derives not from its circular self-justification within the system, but from adjudication by the reader as an external observer, based on the argument's logical coherence and explanatory power.**

This proposal may face a misunderstanding: does it reduce legitimacy to an "appeal to the public"—handing truth over to a majority vote? It must be made clear: no.

There is an essential difference between "external adjudication" and "appeal to the public":

- **Appeal to the public** anchors legitimacy in the number or identity of the adjudicators—"because the majority thinks so, therefore it is true." This is a fallacy.
- **External adjudication** anchors legitimacy in the **rational criteria** according to which the act of adjudication is performed—the adjudicator must judge based on the argument's logical coherence, explanatory power, and correspondence with known facts, not based on personal preference, authority, or numbers.

The difference can be articulated as follows: in an ideal courtroom, the verdict of a jury has legitimacy

not because "there are many of them" or "they are the jury," but because they are required to reach a judgment **based on evidence and law**. Similarly, the reader of this paper is a legitimate adjudicator not because of their number or identity, but because, in reading this paper, they are placed in the position of an adjudicator who must exercise their own rational capacities—they must judge whether the argument of this paper is coherent, whether its terminology is clear, and whether its corollaries have explanatory power. These criteria are not arbitrarily determined by the reader, but are internal requirements of rational argumentation.

Hence, external adjudication is neither the relativism of "whatever you think goes," nor the populism of "the majority decides." It is a matter of shifting the criterion of legitimacy from "who says it" to "how well it is said," while acknowledging that this "how well it is said" must be assessed by a rational adjudicator external to the argumentative system. This is precisely the structure revealed by Gödel's incompleteness theorems—truth is "seen" from outside the system—applied at the meta-theoretical level.

The logical basis for this proposal is this: the ultimate purpose of any meta-theoretical argument is not "self-proof," but persuasion of a rational external adjudicator. The adjudicator stands outside the argumentative system and can simultaneously scrutinize:

- Does the argument follow the logical rules it claims to follow (consistency of the facet of discovery)?
- Is the symbolic expression of the argument clear, unambiguous, and publicly comprehensible (effectiveness of the facet of invention)?
- Do the corollaries derived from the argument accord with known empirical facts (testing of corollaries)?
- Is the overall explanatory power of the argument superior to competing theories (rational criteria of theory choice)?

These criteria are not determined circularly within the argument itself, but are applied by the reader—as an independent external observer. The reader's adjudication is final: if they find that the argument violates non-contradiction, they have the right to reject it; if they find the symbolic expression ambiguous, they have the right to demand clarification; if they find that the corollaries conflict with facts, they have the right to demand revision.

This mechanism is structurally isomorphic to Gödel's incompleteness theorems: the truth-value of the Gödel sentence cannot be proved within the system, but can be "seen" by an observer outside the system (ourselves, equipped with a meta-mathematical perspective). Similarly, the legitimacy of this paper's argument cannot be finally established within the argument itself, but can be adjudicated by the reader outside the argument.

This is not an evasion that "passes the buck" to the reader, but a faithful disclosure of the structure of meta-theoretical legitimacy. Any meta-theory—whether the duality framework of this paper, Quine's revisability of logic, or Dummett's intuitionistic philosophy of logic—is ultimately accepted or rejected not because it has "proved itself," but because it has demonstrated superior explanatory power and coherence before a rational external adjudicator. This paper merely explicitly acknowledges this structure and incorporates it into the argument itself.

7.2.3 Demonstrating Self-Consistency: This Paper as an Instance of Duality

Finally, this paper can and should demonstrate that its own argumentative structure is an instance of the Duality Theorem of Logic. This is not circular argumentation, but a test of self-consistency—if a theory

collapses when applied to itself, it is not self-consistent; if it can account for itself self-consistently, this constitutes powerful support for the theory.

The facet of discovery of this paper: The argument of this paper must strictly follow the fundamental laws of logic—non-contradiction, identity, and causality. If this paper were to contradict itself in the course of its argument (e.g., claiming on the one hand that non-contradiction is inviolable, while on the other hand violating it in its own reasoning), the reader would have the right to reject it immediately. The normative force of this paper derives precisely from its compliance with the rules that it itself reveals. This compliance is not determined by the author's will—the author cannot "choose" to violate non-contradiction in the argument and still expect to be accepted by a rational reader. In this sense, the facet of discovery of this paper is **discovered**: in the course of writing, the author discovered that, for the argument to be valid, it must obey these rules.

The facet of invention of this paper: The organizational structure of this paper, its terminological choices (e.g., "duality," "facet of discovery," "facet of invention," "noetic gene"), its section arrangement, its selection of cases, and its linguistic style—these are all "inventions" by the author to convey the argument effectively. These symbolic organizational forms are not logically necessary: the author could have chosen different terms (e.g., calling the "facet of discovery" the "rule-facet"), could have adjusted the order of sections, could have added or removed cases, could have written in English rather than Chinese. Different symbolic organizational forms can be functionally equivalent or can have differing merits—the author chose the present form based on a comprehensive trade-off between cognitive efficiency (enabling the reader to understand the argument most clearly) and academic conventions (conforming to the norms of scholarly writing). In this sense, the facet of invention of this paper is **invented**: it is the creative choice made by the author within the space of symbols.

The two facets are inseparable: If this paper had only the facet of discovery—that is, only naked logical reasoning, without any symbolic organization—it could not be received by the reader. Logical reasoning must be conveyed through some symbolic form to enter public cognition. If this paper had only the facet of invention—that is, only elegant terminology and structure, but internal reasoning that violates logical rules—it would be a mere rhetorical shell, unable to withstand rational scrutiny. The reason this paper can exist as a qualified piece of academic writing is precisely that it possesses simultaneously a facet of discovery (a logically coherent inferential kernel) and a facet of invention (clear and effective symbolic expression), and the two facets are inseparable.

Thus, this paper achieves a complete self-reference loop: it not only proposes the Duality Theorem of Logic, but also **realizes** this theorem in its own argumentative structure. It demonstrates through its own practice that facing and properly handling self-reference, far from plunging theory into the quagmire of paradox, is the necessary path for a theory to attain meta-level legitimacy.

7.3 Summary: A Methodological Exemplar for Meta-Theory

The work accomplished in this section can be summarized in three steps:

- **Acknowledging self-reference:** Without evasion or concealment. This paper explicitly acknowledges that, as a meta-theoretical paper on the nature of logic, it must itself confront the problem of self-reference.
- **Providing a solution:** By distinguishing intra-systemic self-reference from meta-level self-reference, this paper locates itself in the latter, avoiding the direct threat of Gödelian paradox. Furthermore, by relocating the ultimate criterion of legitimacy from "self-proof within the system" to "adjudication by the external reader," it establishes a structural

foundation for the legitimacy of meta-theory.

- **Demonstrating self-consistency:** This paper applies the duality framework to itself, exhibiting its own facet of discovery (compliance with logical rules) and facet of invention (choice of symbolic expression), thereby proving that this paper is not only a discourse about duality but also an instance of duality.

These three steps jointly demonstrate that the legitimacy of meta-theory resides neither in circular self-justification within the system (which would be a vicious circle), nor in appeal to the number or identity of adjudicators (which would be appeal to the public), but in entrusting the power of adjudication to the rational external observer and trusting them to judge based on the intrinsic quality of the argument—not on extrinsic authority.

These three steps not only safeguard the integrity of this paper's argument, but, more importantly, establish a **reusable methodological exemplar** for meta-theoretical discourse. Future meta-theoretical papers—whether on the philosophy of mathematics, the methodology of science, or moral epistemology—can all draw upon this exemplar: face self-reference squarely, specify the criterion of external adjudication, and demonstrate self-consistency. This may prove to be the most enduring contribution of this paper: it not only answers the question "What is logic?" but also demonstrates "How to legitimately speak about logic."

8. Conclusion

The journey of this paper began with a millennial unanswered question: Is logic discovered, or is it invented? The reason this question has resisted the efforts of countless thinkers is not insufficient evidence or imprecise argumentation, but rather that both sides of the debate have shared an unexamined premise—that logic must be *either* discovered *or* invented. The work of this paper has been to break this either-or framework and to provide a unified account of the nature of logic.

8.1 Core Findings

The core finding of this paper can be encapsulated in a single theorem:

The Duality Theorem of Logic: The logical system simultaneously manifests a dual existence as a facet of discovery and a facet of invention. The facet of discovery is rules as such—the fundamental laws of non-contradiction, identity, and causality. They are constitutive conditions for any stably assertable system, independent of symbolic representation, and endow logic with normativity. The facet of invention is representational forms with symbols as their stable carriers—propositional variables, connectives, quantifiers, inference rule formatting, truth tables, and other symbolic systems. They are the noetic forms through which the rules are written down, transmitted, and evolved, shaped by path-dependence and revisable according to cognitive demands. The two facets are inseparable: without the facet of discovery, normativity is lost; without the facet of invention, operability and evolvability are lost.

The distinctive contribution of this framework is this: it does not seek a "middle ground" between discovery-ism and invention-ism, but directly dissolves the "either-or" mode of questioning. Discovery-ists are right—logic does possess a normative foundation independent of human will. Invention-ists are also right—logical symbolic systems are indeed human creations that evolve over history. Each has glimpsed one facet of the duality of logic but mistaken that facet for the whole.

8.2 Theoretical Contributions

The theoretical contributions of this paper can be articulated at three levels.

First level: Providing a unified framework for the philosophy of logic. The millennial debate between discovery-ism and invention-ism is transformed, within the duality framework, into two facets of a single system. This not only resolves the debate but also reveals the structural reason why the debate remained irresolvable—both sides were using a partial truth to negate another partial truth.

Second level: Providing a dynamical model for the history of logic. Corollary One (the ecological dynamics of logical evolution) interprets the historical transformations of logical symbolic systems as a process of variation and selection: sources of variation include creative innovation, cross-domain transfer, and the driving force of cognitive demands; selective pressures come from representational efficiency, problem-solving power, and systemic compatibility; and the facet of discovery constitutes the inviolable normative boundary of variation. This model provides an operational analytical framework for research in the history of logic.

Third level: Providing architectural principles for logical design in artificial intelligence. Corollary Two (the duality principle for AI logic design) points out that the logical predicament of current large language models stems from "possessing only the facet of invention, lacking the facet of discovery." Based on the Duality Theorem, we have proposed three design principles—hardwiring the facet of discovery, opening the facet of invention, and separating verification from inference—which chart a direction for building AI logic modules that possess both normativity and adaptability.

8.3 Methodological Contribution: The Self-Reference Loop of Meta-Theory

Beyond the substantive theoretical contributions enumerated above, this paper has also accomplished a methodological task: it has demonstrated how a meta-theoretical paper can achieve self-legitimation without falling into paradox.

Any discourse on the nature of logic inevitably confronts the problem of self-reference. This paper has not evaded it, but has squarely achieved a self-reference loop: distinguishing intra-systemic self-reference from meta-level self-reference; relocating the criterion of legitimacy from "self-proof within the system" to "adjudication by the external reader"; and demonstrating this paper's own self-consistency as an instance of duality—the facet of discovery of this paper is the logical rules it follows; the facet of invention is the symbolic expression it employs.

This methodological exemplar not only safeguards the integrity of this paper's argument, but also provides a reusable template for future meta-theoretical discourse. In this sense, this paper not only answers the question "What is logic?" but also demonstrates "How to legitimately speak about logic."

8.4 Directions for Future Research

The work of this paper opens several avenues worthy of further exploration.

Direction One: The applicability of the duality framework to the philosophy of mathematics. Mathematics and logic share similar ontological puzzles: Are mathematical objects "discovered" or "invented"? Does the evolution of mathematical symbolic systems follow a similar duality dynamics? Transferring the framework of this paper to the philosophy of mathematics may yield an equally fruitful unified account.

Direction Two: Empirical testing of the ecological dynamics of logical evolution. Corollary One

yields testable predictions concerning logical symbolic change. Future research may target specific cases from the history of logic (e.g., the transition from syllogistic to predicate logic, the rise of a particular non-classical logic), systematically identify sources of variation, selective pressures, and constraint conditions, and thereby empirically calibrate the dynamical model proposed in this paper.

Direction Three: Engineering implementation of the duality architecture for AI logic. The design principles of Corollary Two have been preliminarily architected in the technical white paper, but considerable work remains to move from architectural design to an operational engineering system. Future research may, based on open-source large language models, attempt to implement a duality logic module with an "axiom verification layer + evolvable symbolic layer," and evaluate its performance advantages relative to purely statistical approaches on standard logical reasoning tasks.

Direction Four: Generalization of the external adjudication mechanism. The "external adjudication" proposed in Section 7 as a mechanism of meta-theoretical legitimacy may have applicability beyond the philosophy of logic. Problems of theory choice in the philosophy of science, and problems of normative foundation in moral epistemology, may both find new analytical tools in this mechanism. Exploration in this direction would extend the methodological contribution of this paper from the domain of logic to the broader study of meta-theory.

8.5 A Final Resonance

Logic is the finest crystallization of human reason. For millennia, it has been both our tool for knowing the world and our mirror for reflecting upon ourselves. The work of this paper has been a meta-level examination of this very mirror: we have discovered that the reason logic can serve both as a strict normative measure and as a continuously evolving response to the expansion of human cognition is precisely that it possesses the dual structure of a facet of discovery and a facet of invention.

The significance of this discovery may extend well beyond the professional boundaries of the philosophy of logic. It suggests a more general worldview: the products of human reason—whether logic, mathematics, scientific theories, or laws, institutions, and cultural narratives—are neither purely objective "discoveries" nor purely arbitrary "inventions." They are noetic fabrics woven together, in the long dialogue between humanity and the world, from the world's immanent rules (facet of discovery) and humanity's own symbolic creations (facet of invention). To understand this duality is to understand the very mode of operation of human reason itself.

The argument of this paper now comes to a close. But as revealed in Section 7, the final power to adjudicate the legitimacy of this paper rests not with its author, but with you, the reader, as external adjudicator. That power of adjudication has now been placed in your hands.

References

Classical Works

- Aristotle. (1993). *Metaphysics* (J. Barnes, Trans.). Oxford University Press. (Original work ca. 350 B.C.E.)
- Aristotle. (1989). *Prior Analytics* (R. Smith, Trans.). Hackett Publishing. (Original work ca. 350 B.C.E.)
- Boole, G. (1854). *An Investigation of the Laws of Thought*. Walton and Maberly.
- Carnap, R. (1937). *The Logical Syntax of Language*. Kegan Paul, Trench, Trubner & Co.
- Frege, G. (1967). Begriffsschrift, a Formula Language, Modeled upon that of Arithmetic, for Pure Thought. In J. van Heijenoort (Ed.), *From Frege to Gödel: A Source Book in Mathematical Logic, 1879–1931* (pp. 1–82). Harvard University Press. (Original work published 1879)
- Gödel, K. (1992). *On Formally Undecidable Propositions of Principia Mathematica and Related Systems* (B. Meltzer, Trans.). Dover Publications. (Original work published 1931)
- Haack, S. (1978). *Philosophy of Logics*. Cambridge University Press.
- Hume, D. (2007). *An Enquiry Concerning Human Understanding*. Oxford University Press. (Original work published 1748)
- Kant, I. (1998). *Critique of Pure Reason* (P. Guyer & A. Wood, Trans.). Cambridge University Press. (Original work published 1781/1787)
- Quine, W. V. O. (1951). Two Dogmas of Empiricism. *The Philosophical Review*, 60(1), 20–43.
- Quine, W. V. O. (1970). *Philosophy of Logic*. Prentice-Hall.
- Russell, B. (1912). *The Problems of Philosophy*. Williams and Norgate.
- Russell, B., & Whitehead, A. N. (1910–1913). *Principia Mathematica* (Vols. 1–3). Cambridge University Press.
- Shapiro, S. (2014). *Varieties of Logic*. Oxford University Press.
- Tarski, A. (1944). The Semantic Conception of Truth and the Foundations of Semantics. *Philosophy and Phenomenological Research*, 4(3), 341–376.
- Wittgenstein, L. (1953). *Philosophical Investigations* (G. E. M. Anscombe, Trans.). Blackwell.
- Wittgenstein, L. (1922). *Tractatus Logico-Philosophicus* (C. K. Ogden, Trans.). Routledge & Kegan Paul.

Contemporary Works in Philosophy of Logic and Cognitive Science

- Dummett, M. (1991). *The Logical Basis of Metaphysics*. Harvard University Press.
- Gabbay, D. M., & Guenther, F. (Eds.). (2001–2018). *Handbook of Philosophical Logic* (2nd ed., Vols. 1–18). Springer.
- Priest, G. (2006). *In Contradiction: A Study of the Transconsistent* (2nd ed.). Oxford University Press.
- Zadeh, L. A. (1965). Fuzzy Sets. *Information and Control*, 8(3), 338–353.

Works from the Author's Theoretical System

- He, G. (2026). *Systemic Ontology: The Ultimate Jurisprudence of Existence*. GitHub/Zenodo Preprint. <https://github.com/wenhe1994/wenhe001>. DOI: <https://doi.org/10.5281/zenodo.19353224>
- He, G. (2026). *Noetic Ecology: A Systemic Theory of Cognition, Culture, and Society*. GitHub/Zenodo Preprint. <https://github.com/wenhe1994/wenhe001>. DOI: <https://doi.org/10.5281/zenodo.19353340>

He, G. (2026). *Civilization Dynamics: Planetary Civilizational Science*. GitHub/Zenodo Preprint. <https://github.com/wenhe1994/wenhe001>. DOI: <https://doi.org/10.5281/zenodo.19352998>

Technical White Paper

He, G. (2026). *The Parliament of Logical Subuniverses: An AGI Alignment Architecture Based on Systemic Dynamics* (Technical White Paper v1.0). GitHub/Zenodo. <https://github.com/wenhe1994/wenhe001>